MIL-C-50072B
30 SEPTEMBER 1964
SUPERSEDING
MIL-C-50072A(MmlC)
18 AUGUST 1961

MILITARY SPECIFICATION

COVER, VOICEMITTER-OUTLET VALVE, C13

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification describes one type of molded cover for the voicemitter-outlet valve of the Ml7 protective mask.

2. APPLICABLE DOCUMENTS

2.1 Government documents The following documents of the issue in effect on date of invitation for bids or request for purposal farm a part of this specification to the extent specified herein:

STANDARDS

FEDERAL

FED-STD-601 – Rubber: Sampling and Testing.

MILITARY

MIL-STD-l05 – Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 –Marking for Shipment and Storage.

DRAWINGS

EDGEWOOD ARSENAL

D5-2-954 – Cover, Voicemitter-Outlet Valve, C13.

PUBLICATIONS

EDGEWOOD ARSENAL DIRECTIVE

No. 618 Ozone Resistant Test of Natural Rubber Material.

(Copies of specifications, standards, drawings, and publicaations required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the cntracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Ratings, Rules and Regulations.

(Application for copies of these freight classification rules should be addressed to the to the Uniform Classification Committee, 202 Union Station, Chicago 6. Ill.)

3. REQUIREMENTS

3.1 Materials.

3.1.1 Rubber compound. “The rubber com-

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pound shall be non-blooming, except for slight wax-type bloom, and shall contain not less than 65 percent, by volume, of new natural rubber, The compound shall be free from lead or mineral sulfates.

3.12 *Antioxidants and accelerators.* The rubber compound shall contain only approved antioxidants and accelerators.

3.1.3 *Sun-checking retardant.* The rubber compound shall contain a sun-checking retardant.

3.1.4 *Copper and manganese inhibitor.* The rubber compound shall contain a copper and manganese inhibitor.

3.1.5 *Coloring materials.* The rubber compound shall be colored by carbon black, as required.

3.2 **Manufacture.** The cover shall be manufactured in one piece by the conventional process of molding under heat and pressure, in accordance with Drawing D5-2-964, in molds approved by Edgewood Arsenal.

3.2.1 *Repair of flaws.* Covers with small flaws may be repaired and accepted, provided there is no change in the formulation or surface continuity.

3.2.2 *Test slabs.* Test slabs, 6 by 6 inches, by 0.075 ± 0.01 inches, shall be made from stock representative of the formulation used for the preproduction sample and from each sublet during regular production of covers. The slabs shall have a cure equivalent to that of the regular production rowers (see 6.3).

3.3 Physical properties Physical properties shall conform to the requirements shown in table I when tested as specified therein.

Property	Unaged	After heat aging	Test paragraph
Average tensile strengh, psi, min	3000	2550	4.7.2.1
Average ultimate elongation, %, min	600		4.7.2.1
Percent increase (max)		20	
Percent decrease (max)		25	
Average tensile stress, psi at 200% elongation	250 to 450	250 to 450	4.7.2.1
Individuals	240	240	
Average tear resistance, lb/in, min			4.7.2.1
Die A or B	300	210	
Die C	225	165	
Average tension set at 75% ultimate elongation, %, max.	20		4.7.2.1
Ozone resistance, min	80		4.7.2.2
Flexibility			4.7.2.8
Units at 75° ± 5°F., min	7.5		
Units retained at −25° ± 5°F., min	4.8		
* Puncture resistance, psi, min	9.5		4.7.2.4

* To be performed on preproduction sample only.

3.4 Preproduction sample. Prior to the start of regular production, a sample cover from each mold cavity and thirty-nine test slabs shall be produced using the same methods, materials, equipment, and rubber formulation as will be employed during regular production. Any change in formulation, operating conditions, or molds shall require a new preproduction sample.

3.5 Workmanship. The cover's inner and outer surfaces shall be smooth and free from imperfections, such as pits, bumps, porus areas, cracks, checks, flash, abrasions, or

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blisters. The rubber shall be compact and free from holes bubbles, and foreign matter. Ridges of mold parting lines on outer surfaces shall not be considered as flash when closely trimmed. Rips, tears, or punctures shall be considered as damage.

4. QUALITY ASSURANCE PROVISIONS

4.1 Respondbility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to insure that supplies and services conform to specified requirements.

4.2 Special provisions.

4.2.1 *Alternative inspection. The* supplier may utilize any alternative inspection procedure which will provide equal or better assurance of quality by submitting a written proposal with justification and obtaining written approval from the Government prior to instituting the procedure. In case of dispute, the procedures of this specification will govern.

4.2.2 Objective *evidence.* The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2.3 *Inspection substation.* If the supplier elects to establish inspection substations for the purpose of conducting inspection for certain listed defects, the Acceptable Quality Levels (AQL's) for each inspection substation shall be equal to the AQL which is the next numerically lower value than the AQL specified for the total listed defects. The requirements of MIL-STD-105 are applicable to each substation and to the end item.

4.3 Preproduction sample inspection.

493.1 *Examination. The* sample covers shall be inspected for all requirements of the drawings and of this specification.

4.3.2 Test. Test slabs produced in accordance with 8.4 (see 3.2.2) shall be tested in accordance with 4.7.2.

4.3.3 *Acceptance/rejection* criteria. The preproduction sample shall meet the examination and tests specified by 4.3.1 and 4.3.2 to be acceptable. The supplier shall obtain written approval from the contracting officer before proceeding with regular production.

4.4 Lotting. A regular production lot shall consist of the covers produced in sequence during one month, as indicated by the dots in the medallion on the cover. A sublet shall consist of the covers produced in not more than one work week from one batch of material (see 6.6). A new lot shall be established following the occurrence of any event affecting the homogeneity of the lot or the use of an alternate approved formulation. Acceptably repaired covers from any accepted parent lot may be incorporated into and assume the lot number of current production.

4.5 sampling.

4.5.1 *For examination.* Sampling shall be conducted in accordance with MIL-STD-105.

4.5.2 *Fur destrutive tests.* Thirty-nine test slabs (see 3.2.2) shall be submitted with each sublet for testing.

4.6 Inspection provisions.

4.6.1 *For examination. The* sample covers from regular production lots shall be examined in accordance with the classification

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of defects and with MIL-STD-105.

4.6.2 For destructive tests. Sample test slabs shall be tested in accordance with 4.7.2. The acceptance number is 0.

4.63 Classification of defects.

4.63.1 Cover, voicemitter-outlet valve (Dwg D5-2-954).

Categories	Defects
critical :	None defined
Major:	AQL 1.0 percent defective.
101	Workmanship (see 3.6)
102	Fit incorrect
103	Rivet-hole diameter or alignment incorrect
104	Anti-tear slit incorrect or incorrectly located.
106	Anti-tear slit not clean cut or free from nicks or flash
Minor:	AQL 2.5 percent defective
201	Thickness incorrect
202	Marking missing, illegible, or incorrect

4.7 Tests.

4.7.1 *Test specimens,* Specimens shall not be buffed or stretched before testing. Twenty specimens, two from each of 10 test slabs, shall be taken for tensile strength, ultimate elongation, and tensile stress testing. Ten specimens shall be tested before aging and 10 shall be tested after aging. Six specimens, two from each of three test slabs, shall be taken, without aging, for tension-set testing. Twenty specimens, two from each of 10 test slabs, shall be taken for tear-resistance testing. Ten specimens shall be tested before aging and 10 shall be tested after aging. Twelve specimens, two from each of six test slabs, shall be taken for ozone-resistance testing. Eight specimens, two from each of four test slabs, shall be taken for flexibility testing. Six specimens, one from each of six test slabs, shall be taken for puncture-resistance testing. The test specimens selected for heat aging shall be aged in an air oven at a temperature of 158° ± 1.8°F. for a continuous period of 168 ± 1 hours or at a temperature of 212° ± 1.8°F. for 24 ± 1 continuous hours, as specified in FED-STD-601 (see 6.7).

4.7.2 *Physical properties* test.

4.792.1 *Tensile strength, ultimate elongation, tensile stress,* tear resistance, and tension set. Tensile strength, ultimate elongation, tensile stress, tear resistance, and tendon set shall be determined as specified in FED-STD-601. The angle-type or crescent type die shall be used for tear-resistance testing.

4.7.2.2 Ozone resistance. Ozone resistance shall be determnined in accordance with Directive No. 618.

4.7.2.3 Flexibility. Flexibility shall be determined using Edgewood Arsenal Q37 Natural Rubber Flexibility Tester. Flexibility shall be read on the grid of the apparatus. Four specimens shall be tested at 75° ±5°F. and four specimens shall be tested at –25° ±5°F, after being conditioned for 72 hours at each temperature.

4.7.24 Puncture resistance. Resistance to hydrostatic pressure shall be determined using Edgewood Arsenal Q38 Natural Rubber Puncture Tester.

5. PREPARATION FOR DELIVERY (For repair parts packaging, see 6.8).

5.1 Packaging, leveI C. Covers shall be packaged to afford adequate protection from the supply source to the first receiving activity for immediate use.

5.2 Packing, level C. Covers, packaged as specified in 5.1, shall be packed to insure carrier acceptance and safe delivery at first domestic destination. Containers shall comply with Uniform Freight Classification rules or regulations of other carriers applicable to the mode of transportation.

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5.3 Marking. In addition to any special marking required by the contract or order, that packages and shipping containers shall be marked in accordance with MIL-STD-129. Marking shall include the date of cure.

6. NOTES

6.1 Intended use. This specification describes a protective cover intended to be used. on the C13 voicemitter-outlet valve.

6.2 Ordering data. Procurement documents should specify the title, number, and date of this specification.

6.3 Equivalent cure. Slabs and covers of the same stock and gage are considered to have equivalent cures or the same state of valcanization when:

(a) Their free-sulfur contents do not vary more than 10 percent

(b) Their average tensile-stress values at any given elongation (other than ultimate elongation) do not vary more than ± 2.5 percent.

(c) Their curing times can be equated in the following equation:

$$\frac{T_1 - T_2}{10}$$

$$t_2 = t_1 \times 1.6$$

where: t_1 = minutes of cure at temperature T_1,

t_2 = minutes of cure at temperature T_2,

T_1 = temperature (°F.) of cure for time t_1, and

T_2 = temperature (°F.) of cure for time t_2.

For convenience, the nomograph in figure 1 may be used instead of the mathemtical computation time-temperaure relationship in determining equivalent cure.

6.4 Major change. The term applies to any change in formulae, processing, or curing of currently approved ruber stocks. The major change should be considered effected when one or more of the following conditions occur:

(a) Component of formulae, are varied in excess of the limitations stated in the manufacturers' tolerance (see 6.5).

(b) The addition or deletion of one or more components of f formulae.

(c) Substitutes for one or more components of formulae without advance approval of Edgewood Arsenal.

(d) Deviations from the submitted and approved factory mold cure exceeding ± 5°F. or –lO°F., even though equivalent cure is attained.

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EQUIVALENT CURE TIME

FOR

NATURAL RUBBER

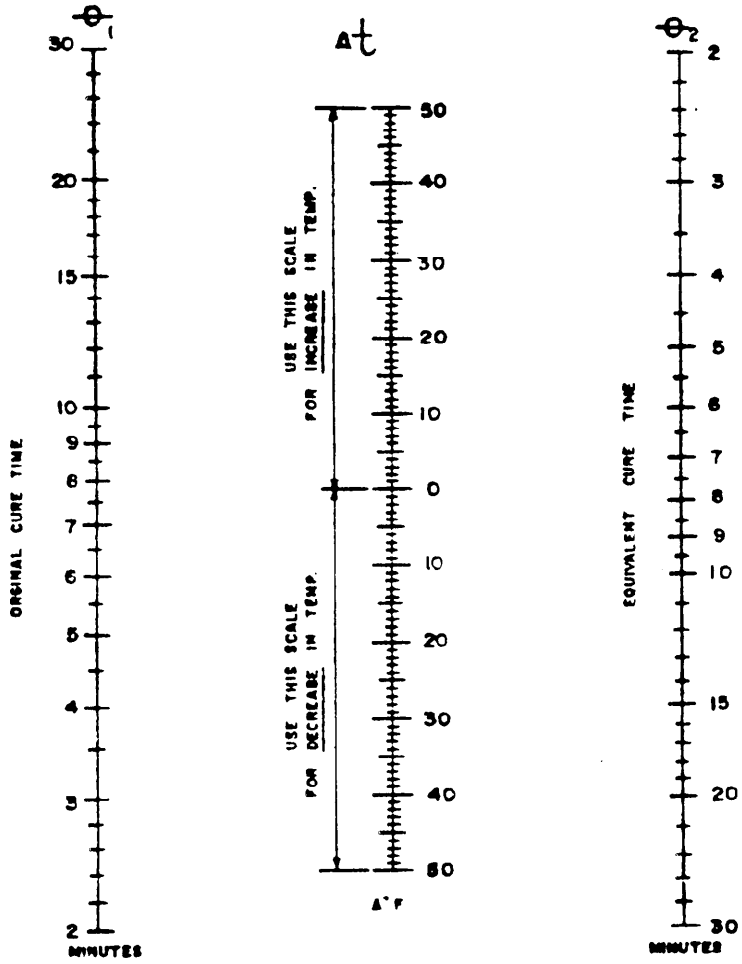


Figure 1.

USING A STRAIGHT EDGE, CONNECT THE ORIGINAL CURE TIME ON θ_1 SCALE WITH THE CHANGE OF CURE TEMPERATURE ON Δt SCALE. EXTEND THE LINE UNTIL IT INTERSECTS θ_2 SCALE AT THE EQUIVALENT CURE TIME.

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(e) Use of mold lubricants, dusting powders, or wet sprays not used in preparation or cure of the original samples submitted for approval.

6.5 Manufacturer's tolerances. The manufacturer should submit to Edgewood Arsenal the variations in amounts used, or tolerances, for accelerators, plasticizers, and softeners. These tolerances are permitted to enable the manufacturer to produce stock which will consistently comply with this specification and which will process satisfactorily in the plant. Components of formulae may be submitted on the basis of percentage of stock or number of parts of rubber, and the manufacturer's tolerances recorded in the same units.

6.6 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process and subjected to some physical mixing operation intended to make the final product substantially uniform.

6.7 Aging. Aging at 212° ± 1.8°F. for a continuous period of 24 hours is slightly more severe on rubber than aging at 158° ±1.8°F. for 168 consecutive hours; therefore, specimens that fail to meet requirements when aged at the higher temperature may be discarded and replaced by new specimens aged at 158° ±1.8°F. for 168 consecutive hours.

6.8 Repair parts packaging. when repair **parts** packaging and packing is required for items authorized for stocking as repair parts, the repair parts packaging data sheet covering the outlet valve cover should be used. (Copies of repair parts packaging data sheets may be obtained from the Director of Engineering and Industrial Services, ATTN: Chief, Design Branch, US Army Edgewood Arsenal, Edgewood Arsenal, Md 21010.)

6.9 International interest. Certain provisions of this specification are the subject of ABC international standardization agreement. When amendment, revision, or cancellation of this specification is proposed, the departmental custodians will inform their respective Departmental Standardization Offices so that appropriate action may be taken respecting the international agreement concerned.

Custodians:
Army-MU
Navy-SH
Air Force-84

Preparing l ctivity:
Army-MU (EA)
Project NO. 4240-0058

International interest (see 6.9)

Review activities:
Army-MD
Navy-SH

User activities:
Navy—MC
Air Force-84

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION

ORGANIZATION (Of submitter)	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity)	DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-8279

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U. S. ARMY MUNITIONS COMMAND
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PLATE NO. 15418 (BACK)

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